2.4 Differentiating the Product and the Quotient of Two Functions; Higher-Order Derivatives

Name _____ Period ____ Date _____

AP Calculus 2.4 (30-minute plan) · Student sheet

I can…

- use the Product Rule.
- use the Quotient Rule and the Power Rule for negative exponents.
- find second and higher-order derivatives, including acceleration.

Vocabulary · fill in the missing word

Product Rule — $(fg)' =$ _____

Quotient Rule — $\left(\frac{f}{g}\right)' =$ _____, $g \neq 0$

Second derivative — _____

Part 1 · Notes and examples (fill in each note, then work the example)

Product Rule: $(fg)' = f'g +$ _____.

Example 1

Find $f'(x)$ for $f(x) = (x^2 + 1)(3x - 2)$.

Quotient Rule: $(f/g)' = (f'g - fg')/$_____.

Example 2

Find $f'(x)$ for $f(x) = \frac{x^2 + 1}{2x - 1}$.

Power Rule, negative exponents: $d/dx\ x^{-n} =$ _____.

Higher-order derivatives: f'' is the derivative of f'; if $s(t)$ is position, $s''(t)$ is the _____.

Example 3

a. $f(x) = 3x^4 - 2x^3 + x$. Find $f''(x)$.

b. $s(t) = t^3 - 6t^2$. Find the acceleration at $t = 3$.

Part 2 · Partner practice

Partner practice

(AP® Practice) If $f(x) = x^3 + 4x^2 - 2x$, then $f''(1) =$

(A) 12 (B) 14 (C) 9 (D) 6